

RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty : Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title : OML 751 Testing of Materials
Year and Branch : IV year Civil, EEE and ECE
Topics Covered : Unit -II: Mechanical Testing –Rockwell
Hardness Test
Name of the Laboratory : Strength-of-Materials lab
Website Link : <http://sm-nitk.vlabs.ac.in/exp20/index.html>
Date : 26.08.2020
No.of Students participated : 62
Assignment given if any : -
(If Yes share the uploaded
folder link)

Demonstration of Rockwell Hardness Test

OBJECTIVE

To find the Rockwell hardness number of mild steel, cast iron, brass, aluminium and spring steel etc.

Hardness represents the resistance of material surface to abrasion, scratching and cutting, hardness after gives clear indication of strength. In all hardness tests, a define force is mechanically applied on the piece, varies in size and shape for different tests. Common indenters are made of hardened steel or diamond. Rockwell hardness tester presents direct reading of hardness number on a dial provided with the machine. Principally this testing is similar to brinell hardness testing.

The students can able to know the different type of Rockwell test procedure, specifications and procedure.

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO2 Analyze engineering components using various mechanical testing procedure.	2	2	1	2	2	2	2	1	2

ROCKWELL HARDNESS TEST

STEP 3 Based on the material the major load is set & the motor is switched ON. The hand lever is kept in load position. The reading is noted down using B-scale or C-scale based on indenter used.

TRIAL : 1

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ROCKWELL HARDNESS TEST

STEP 4 Two trials is noted for all the four specimens and the average of the two trials gives the final Rockwell Hardness number.

Trial	Material	Total Load P (Kg)	Scale Used	Rockwell Hardness number (RHN)
1	Brass	100	B	68
2				38

Average Rockwell Hardness number = 53.00

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